

Caio Valerio Goulart Correia

AI Specialist · Data Engineer · Backend Developer

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AI Specialist and Backend/Data Engineer with 9+ years of technical experience, currently focused on LLM training, evaluation, and alignment for frontier models. Hands-on with SFT and preference optimization (DPO/RLHF) and parameter-efficient fine-tuning (LoRA/QLoRA) using PyTorch, Hugging Face, Unsloth, and Axolotl, plus local LLM serving and agentic tooling (vLLM, SGLang, llama.cpp, MCP) on dedicated NVIDIA hardware. Builds data-intensive products end to end: geospatial pipelines processing 100M+ point clouds, an HR platform serving 200+ employees under strict PII controls (LGPD), and RAG-based compliance auditing. Strong across Python, TypeScript/Node.js, and PostgreSQL, with cloud/DevOps delivery on AWS, Docker, Kubernetes, and GitHub Actions. Mechanical engineering background with a track record of turning complex industrial processes into reliable, measurable software. Fluent English (C1), experienced with international, remote-first teams.

PROFESSIONAL EXPERIENCE

AI Specialist (Contract)

micro1 • Aug 2026 – Sep 2026 • Remote

- Trained and evaluated LLMs for frontier-model programs, running Supervised Fine-Tuning (SFT) and preference optimization (DPO/RLHF) with PyTorch, Hugging Face PEFT (LoRA/QLoRA), Unsloth, and Axolotl.
- Ran adversarial testing (red teaming) to surface hallucinations, logic flaws, and bias, feeding corrections back into training datasets.
- Reviewed and benchmarked AI-generated code across Python, TypeScript, Node.js, FastAPI, React, Next.js, and JavaFX — assessing data-structure choices, execution behavior, and complex API integrations.
- Deployed local LLM serving and agent harnesses on Windows/Linux (vLLM, SGLang, llama.cpp) on dedicated NVIDIA hardware (L4 GPU, 24GB VRAM), plus MCP tool servers for retrieval and graph tools; measured latency/throughput with custom evaluation harnesses.
- Curated high-quality training and synthetic datasets (Self-Instruct, Evol-Instruct) with automated validation for dataset integrity, schema compliance, and model alignment.

Full-Stack Developer & DevOps Engineer ABZ Serviços • Mar 2025 – Present • Rio das Ostras, RJ

- Architected and shipped EmployeeHub (portal.groupabz.com), an HR/operations portal used by 200+ employees — personnel records, crew management, e-Social, vacation, reimbursement, and ACL — on Next.js 15, TypeScript, and Supabase (PostgreSQL with Row-Level Security) on Vercel; reduced HR manual workload by ~40% and approval time by ~95%.
- Delivered the ABZ product suite around the portal: PontoFlow (bilingual offshore timesheets with manager approval, field annotations, and deadline locks enforced in UI and RLS), Invoice ABZ (FastAPI/SQLAlchemy engine turning payroll and charge-rate spreadsheets into client invoices), and Ticket-Manager (support automation via Microsoft Graph).
- Led the cloud migration of legacy on-premise Windows Servers (2012–2025), reaching 80% completion with zero production downtime.
- Deployed a WireGuard-based VPN for 50+ remote employees, sharply reducing connectivity support tickets and failures.
- Implemented end-to-end CI/CD with GitHub Actions and Docker; administered Proxmox virtualization and AWS S3.

Python Developer & Spatial Data Engineer BRTech3D • Mar 2024 – Mar 2025 • Macaé, RJ

- Built high-performance Python pipelines (NumPy, SciPy, Open3D) automating geospatial data processing for dense point clouds (100M+ points), increasing processing speed by 40% and improving data compression across dozens of projects.
- Modeled and optimized PostgreSQL databases with the PostGIS extension for complex spatial queries, integrated via REST APIs.

Topographical Survey Operator III (Laser Scanning) Laser Master Engineering Services • Aug 2023 – Mar 2024

- Operated high-precision 3D laser scanning equipment (Leica) in complex industrial and archaeological environments.
- Led field campaign planning and technical documentation, ensuring millimeter-level precision and data backup redundancy for large-scale data collection.

Topographical Survey Operator II BRTech3D • Dec 2022 – Aug 2023

- Executed field data collection using Leica TLS equipment and introduced Quality Assurance (QA) processes for outlier cleaning and multiple scan registration.

Technical Documentation & Support Specialist SmartStore / Jackkk Connection • Mar 2021 – Dec 2022

- Produced 50+ bilingual technical manuals and system specifications for international clients.
- Provided advanced pre-sales support and detailed hardware/software architecture mapping for complex business proposals.

Training Specialist & Automation Technician Jackkk Connection Comercial • May 2017 – Mar 2021

- Coordinated and delivered technical training for over 500 professionals in PLCs, hydraulics, and industrial automation.
- Developed educational simulations and managed B2B technical sales, building direct relationships with international suppliers.

Technical Team Lead & Specialized Support Diesel Line Cambuí • Aug 2016 – May 2017

- Led specialized field teams in the maintenance of large-scale engines and turbines (CAT, Cummins, MAK).
- Managed English-language technical support tickets for international clients and formally documented failure analysis processes.

Apprentice Technician Diesel Line Cambuí • Oct 2015 – Aug 2016

- Began career providing support to commercial processes, CRM management, and strict application of compliance procedures for diesel and gas systems maintenance.

RELEVANT PROJECTS

CloudSec & FinOps Compliance Auditor Portfolio Product • 2026

- **Live:** cloudsec-finops-auditor.vercel.app · **Repo:** github.com/Caiolino00/cloudsec-finops-auditor
- Production-oriented compliance auditor that evaluates cloud architecture scenarios against CIS / SOC 2 / FinOps-style policies, returning structured risk, cost-impact, citations, and remediation steps.
- Stack: Next.js (App Router) + TypeScript + Tailwind on Vercel; hybrid RAG (BM25 + TF-IDF + Reciprocal Rank Fusion) over versioned policies/*.md; structured Gemini audits (gemini-3.8-flash → gemini-3.6-flash fallback with 503 retry); bilingual EN/PT UI; Vitest unit tests and CI evaluation pipeline.

Portal ABZ (EmployeeHub) ABZ Serviços · Internal HR / Operations Portal · Live

- **Live:** portal.groupabz.com · **Repo:** github.com/Caiolino00/EmployeeHub
- Built the ABZ Group corporate portal (EmployeeHub / Painel ABZ) covering personnel records (DP), crew management, Man Schedule, e-Social, vacation, reimbursement, and ACL — the operational source of truth for people and offshore schedules.
- Stack: Next.js 15 + TypeScript + Supabase (PostgreSQL/RLS) on Vercel; payroll derived from portal embarkation data with WK Radar sync, printable payslips, and audited, reversible schedule edits.
- Integrates with the ABZ suite: identity/collaborator data feeds Timesheet (PontoFlow) sync paths; payroll/charge-rate outputs feed Invoice generation for clients.

PontoFlow (Timesheet Manager) ABZ Group · Timesheet + mobile · Standalone (EmployeeHub-ready)

- **Live:** ponto-flow.vercel.app · **Repo:** github.com/Caiolino000/PontoFlow
- Delivered PontoFlow, a bilingual (pt-BR/en-GB) timesheet system for offshore crew with manager approval, field-level annotations, and monthly deadline lock enforced in UI and RLS.
- Stack: Next.js 15 + TypeScript + Supabase + Nodemailer; email/cron reminders (T-7...T); mobile companion; multi-tenant admin (invites, per-tenant SMTP).
- Built standalone with EmployeeHub-ready sync (HMAC-authenticated import/export) so collaborator identity and timesheet periods can align with the Portal ABZ suite.

Invoice ABZ (Payroll → Invoice) ABZ Group / ABZ International · Finance automation (private repo)

- Built a multi-company Invoice engine that turns payroll/charge-rate spreadsheets into client invoices (FMS, OMEGA, Jifmar vessels) with automatic client detection and parser routing.
- Stack: FastAPI + SQLAlchemy + SQLite + OpenPyXL; preview → invoice number/date → per-person invoices or ZIP; isolated dev/prod databases.
- Closes the ABZ financial loop: Portal/folha and charge-rate data become standardized invoices for ABZ Group and ABZ International without rewriting code per client.

Model Context Protocol (MCP) / Knowledge Graph

Personal / Local AI stack

- **Repo:** github.com/Caiolino000/sglang-commander
- Built MCP tool servers over a local knowledge graph (Python + SQLite/FTS indexing) so agents retrieve structured neighborhoods instead of stuffing raw context into prompts.
- Wired tool-calling loops to OpenAI-compatible local serving (vLLM / SGLang) and llama.cpp for low-latency/offline paths; custom harnesses for multi-step agents, retries, and tool-success eval. Includes SGLang Commander (repo link), a desktop + web GUI to manage SGLang inference servers.

TECHNICAL SKILLS

AI & LLM Engineering

PyTorch, Hugging Face (Transformers, PEFT, TRL), LoRA/QLoRA, Unsloth, Axolotl, DeepSpeed, SFT, DPO, RLHF, LLM evaluation & red teaming, synthetic data generation (Self-Instruct, Evol-Instruct), RAG, LangChain, LlamaIndex, vLLM, SGLang, llama.cpp, NVIDIA Triton Inference Server, Model Context Protocol (MCP), agent harnesses / tool calling.

Backend

Python (FastAPI, SQLAlchemy, NumPy, SciPy, Open3D), Node.js, Express, TypeScript, Java (Spring Boot), Go (automation scripting), REST, GraphQL, JWT.

Frontend

Next.js 15 (App Router), React, Tailwind CSS.

Data & Databases

PostgreSQL, PostGIS, Supabase (Row-Level Security), SQLite (FTS5), NoSQL, ETL/data pipelines, data lakes, geospatial & point-cloud processing.

Cloud & DevOps

AWS (S3), Docker, Kubernetes, Terraform, GitHub Actions (CI/CD), Vercel, Cloudflare, Proxmox, WireGuard VPN, Linux (Ubuntu/CentOS), Windows Server, PowerShell, Microsoft Graph API.

Testing & QA

Vitest, CI evaluation pipelines, automated dataset validation, schema compliance checks, code review.

EDUCATION

B.S. in Mechanical Engineering (in progress) UniFatecie • May 2024 – May 2027 (expected)

Technical Diploma in Industrial Automation ETP Escola Técnica Brasil • Jan 2014 – Jan 2016

LANGUAGES

Portuguese:

Native |

English:

Fluent (C1 — technical and conversational) |

Spanish / Italian:

Professional working proficiency, advanced technical reading.